Design Proposal for MIME Capstone Design
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Team #618
Musical Bot Encore

Musical Bot Encore: The Modular Keyboard-Playing Robot

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1. Introduction and Project Scope

Engineering Expo is one of the most prominent outreach events organized by the College of Engineering. For many K-12 attendees, this event represents their first hands-on experience with a robot built by a student. The historic GuitarBot exhibit has served this role admirably in the past; however, its age, wear, and tear on sound quality have diminished its effectiveness. Musical Bot Encore aims to replace this exhibit with a novel electromechanical system that mounts on a household keyboard to play songs autonomously. Our goal is to create an exhibit that is reliable, portable, visually interesting, and able to withstand the rigors of multiple trips to Engineering Expo to captivate K-12 attendees and inspire them to become future College of Engineering students who can appreciate the intersection of mechanical engineering, electrical engineering, and software engineering.

The system takes pre-selected digital music and translates it into mechanical key presses. A software pipeline decodes the music data, an Arduino controller manages the schedule of note presses, driver circuitry controls the required currents, and an array of solenoids presses the keys through a non-invasive frame. The system's use of an actual keyboard to play music instead of a display screen helps to preserve the cause-and-effect connection between the system's actions and its demonstration of mechanical engineering principles that makes the system more memorable to attendees while still being replicable with simple electronics, 3D-printed parts, and open-source design tools.

1.1 Scope Definition

The scope of this proposal is to design a robot to play a keyboard, not a production-quality autonomous music system. The scope of this project includes designing a modular robot to play complete songs without human intervention after initialization, to mount on a keyboard without modification to the keyboard, to assemble quickly to allow for multiple demonstrations at public events such as Engineering Expo, to be portable, visually interesting, and to be built on a student budget. The design should also be extensible to allow future teams to expand the repertoire of songs beyond the initial set of songs chosen for demonstration and perhaps to incorporate additional features such as more keys or sustain pedal support in the future.

1.2 Stakeholders and Customer Requirements

The major stakeholders for this project include Dr. Sarah Oman as the project sponsor, MIME

Capstone instructional staff, Oregon State University's College of Engineering, Engineering Expo staff, and future capstone teams that may maintain or expand the system. The public would include students in grades K-12 and their families. Therefore, the design would need to be technically sound yet accessible to this group. The customer requirements that the team has determined include electronically driven autonomous playing, correct note play, nice sound, aesthetically creative exterior design, easy and interactive design, wide note and dynamic capability, non-invasive securing system, portability, and fast setup. These requirements would be quantitatively defined as shown in Table 1. For this project, a successful song run would be considered one that runs from start to finish without any failure that would negate the functionality of the system. It would not require any human input after the start command.

Table 1. Customer requirements translated into engineering specifications.

Specification	Target	Why It Matters
Autonomous operation	>=95% successful song runs over 20 trials; 0 human inputs after start	Primary reliability metric for Engineering Expo use
Song library	At least 3 playable songs	Keeps the exhibit varied and repeatable
Material reuse	>=50% recycled materials by part count	Supports sustainability goals
Setup and portability	<10 min setup and <100 lb total system mass	Enables transport and rapid event deployment
Mounting method	Non-invasive attachment to different home keyboards	Protects equipment and broadens compatibility
Performance quality	Correct note playback with pleasant sound and visible actuation	Maintains technical credibility and audience appeal

1.3 Background and Motivation

Past research on robotic musicianship has revealed that a computerized music performance system is a viable educational tool, yet also poses a complex engineering problem. The development of robotic percussion and keyboard systems has clearly shown that proper note triggering is achievable, yet making a convincing music performance is more than simply striking the right note [1], [2]. The study on robotic piano music performances revealed that timing, articulation, and dynamic variations significantly affect a music performance, making it sound more human or machine-like [3]-[5]. This means that for Musical Bot Encore, the project does not require concert-level expressiveness to succeed as an outreach exhibit, yet we need to be able to control timing and striking intensity well enough to avoid a machine-like sound.

2. Constraints, Risks, and Assumptions

2.1 Constraints

The key constraints in the project are:

- The system should be ready for Expo within the constraints of the MIME 497/498 timeline.
- The system should be compatible with a student budget of about $1,000.
- The system should be non-invasively attachable to a home keyboard, requiring no permanent modification to the keyboard.
- At least 50% of parts should be recycled or reused.
- The system should be portable, require less than 10 minutes to set up, and weigh less than 100 lb.

2.2 Risks

The major technical risk here is that there may be insufficient actuation speed or consistency. If the switching hardware cannot act fast enough for successive note events, the robot may not be able to play the notes. This has already been a risk in the prototype design, as a MOSFET driver board was assumed to be available, but was actually a relay board. Testing showed that relay hardware could not act fast enough for certain passages. This risk will be mitigated in the ongoing design by the use of MOSFET switches, PWM control, and a stepwise approach from a single key to a small cluster before integration.

A second major technical risk is that there may be misalignment between the solenoid plungers and the keys. This could cause noise, inconsistency, and side loading. This risk can be mitigated through the use of adjustable mounting arms, brackets, contact pads, and a stepwise approach from a single key before integration.

A third major technical risk here is that there may be a tendency for the design to get too complex for a capstone project to build and verify in a timely fashion. While a full keyboard, elaborate exterior design, and sustain pedal features would be nice additions, there may be a tendency for these features to cloud the main design goals if pursued too aggressively. Therefore, the design will focus on a subset for the capstone while allowing for expansion in the future after reliable functionality has been demonstrated.

2.3 Assumptions

It is assumed that a home keyboard will be available that has sufficient key geometry for mounting, that solenoids can be obtained that have sufficient force and actuation time for the design, that framing materials can be reused without safety issues, and that the software design can be organized so that MIDI note events can be converted to Arduino commands without

human intervention during a song run.

3. Design Process and Alternatives

The concept was developed as a result of a structured design process as opposed to personal preference. The steps taken to develop the concept included translating requirements into specifications, building proof-of-concept hardware to identify key risks, and developing multiple system-level concepts and then selecting based on weighted criteria to match customer requirements.

3.1 Requirements Translation and Benchmarking

The starting point in the design process was to take customer requirements like portability, pleasant sound, and variety in notes and translate these into requirements that would drive the design. Benchmarks were also considered to identify how to make it successful in a public display environment. The requirements translated into how fast it should take to set up, how engaging the mechanism should be, and how reliable it should function so that users are not discouraged by failures. It was also realized at this stage that it was an integrated mechatronics problem involving mounting, electrical switching, timing, and user interface.

3.2 Proof-of-Concept Development

The development of a proof of concept was a key activity in the design process. The initial proof of concept was developed by placing a cardboard frame over a keyboard. It was able to play notes and even a rough version of “Mary Had a Little Lamb.” It was realized at this stage that there were significant integration challenges. The initially proposed hardware was not appropriate for the tempo required in playing. The steps taken from this point in the design process were rational and risk retirement-based. These steps included drawing a wiring diagram for one solenoid, designing a solenoid mount and interface to the keyboard, building a housing for MOSFET, Arduino, PWM, and voltage splitting, fixing bugs in the MIDI reading code, streamlining the Python to Arduino translation, and organizing the code to make it expandable to 61 keys.

3.3 Concept Generation

There were three main concepts that were analyzed, and they included the following:

For Concept A, the approach was based on the use of an array of solenoids, where each key in the range would have an actuator assigned to it [6].

For Concept B, the approach was to have fewer actuators moving over many keys via the use of a moving carriage or rail-based approach.

For Concept C, the approach was the use of linkage or cam-based multiplexing.

All the options had the capability of playing keys, but the issue was determining the best approach in terms of meeting the requirements of the project's priorities. Concept A offered the best approach in terms of the autonomous multi-note play, where each key had an actuator assigned to it without the need for the actuator moving in the lateral direction.

Concept B reduced actuator count, but introduced carriage motion and control complexity that increased schedule risk. For Concept C, the approach offered the advantages of fewer actuators, but the backlash, tuning, and lack of independent control of the playing of notes offered challenges in accurate note execution.

Based on the requirements of the customers, the key factors included the playing of the notes correctly, the sound being pleasing, autonomous, portable, and expandable, where the simplest approach in terms of the number of parts was not necessarily the best.

3.4 Downselect Criteria and Results

The downselect criteria were focused on the key aspects of the project, where the key factors included the autonomous approach, the range of notes, the dynamics, the non-invasive approach, the portability, the build complexity, the cost, and the expandability of the final product. The representative weighted decision matrix used for the downselect is shown in Table 2.

Based on the requirements, the final product had to have the capability of playing multiple notes, where the timing was crucial in terms of the public demonstrating the capabilities of the product, where the credibility of the product would only last for a short time if the playing of the notes was not done well.

Table 2. Representative weighted decision matrix for the major system concepts.

Criterion	Weight	Concept A: Fixed Array	Concept B: Moving Carriage	Concept C: Linkage/Cam
Autonomous reliability	0.25	5	2	2
Note range and dynamics potential	0.20	4	3	2
Non-invasive mounting and portability	0.15	4	3	3
Build complexity risk	0.15	3	2	2
Cost and fabrication burden	0.10	3	4	4
Future expandability	0.15	5	3	2
Weighted total	1.00	4.15	2.70	2.35

Concept A was given the highest weighted score since it was the most suitable solution to the project's most significant requirements, even though it was not the most cost-efficient solution. The main disadvantage of this solution is that it increases the number of channels, leading to increased wiring and power requirements. However, this disadvantage is feasible in a staged capstone project, while the other solutions compromise the project timeline and multi-note capability by reducing the number of hardware components used.

4. Proposed Design Solution

4.1 System Overview

The proposed design solution is a modular fixed array keyboard playing robot. The song data is preprocessed on a host computer and then transmitted to a Musical Bot Encore controller based on an Arduino board. The controller schedules the note times and sends PWM signals to the MOSFET driver electronics, which switch the current to the solenoids. The solenoids are mounted on adjustable arms connected to a non-invasive structure positioned above the keyboard. This design solution decouples the software timing, power switching, and actuation to enable independent subsystems to be improved without requiring changes to the entire system.

4.2 Mechanical Architecture

The proposed solution's mechanical architecture is based on a lightweight structural bridge that connects to a home keyboard without any need to permanently modify the keyboard. Adjustable arms or brackets hold the solenoids above the corresponding keyboard key to enable tuning of the positions during assembly. This design supports compatibility with multiple home-keyboard geometries while also enabling rapid setup and disassembly, allowing many noncritical support parts to be made from recycled parts.

Each of the actuation channels conforms to a uniform hardware arrangement, with a bracket, solenoid, plunger guide, and key contact interface. This modularity improves serviceability, as well as expansion, because the key station can be tested, adjusted, and then repeated across the range of keys or keyboard layouts the team elects to implement. Compliant contact pads minimize noise and local stress at the key interface, while adjustability helps maintain repeatable travel and ordered motion during demonstrations [7].

4.3 Electrical and Control Architecture

In the electrical architecture, the project utilizes the Arduino microcontroller to schedule the electrical signals. This microcontroller works with MOSFET electronics, which serve as the drivers of the electrical signals. This is necessary because the microcontroller sends the timing signals, but the solenoids need more current than the microcontroller's pins can provide. This change from the original relay hardware represents an important correction identified during prototype development. In the electrical architecture, the power supply is centralized, fused, and encased. This is necessary because the project will transport the keyboard to be used during demonstrations at public events. As the number of actuated keys increases, total current demand will also increase, so power distribution and integration testing will be critical during system expansion and public demonstrations.

4.4 Software Pipeline and Performance Strategy

The software pipeline involves the selection of the song, the processing of the MIDI events, and the eventual transfer of the events to the Arduino for playing the song. Some of the important tasks include fixing the bug in the script used for reading the MIDI, as well as the transfer of the Python script to the Arduino, which is crucial for the autonomous playing of the songs rather than housekeeping tasks. Using PWM is a realistic approach towards incorporating dynamics, although the expressive nuances are not necessarily important for the success of the program, the control over the energy of the strikes, as well as the duration, is important in ensuring that the playing does not appear flat or robotic over time [8]. The structure of the code, although long in the current form, is important in the long run for incorporating the keys as well as the sustain pedals, although they would have to follow the successful demonstration of the autonomous playing of the songs.

4.5 Advantages and Disadvantages Relative to Customer Requirements

The selected design has five advantages compared to the customer requirements. First, the design can be electronically controlled and operated autonomously. Second, the fixed-array design naturally supports overlapping notes compared to the moving or multiplex designs. Third, the non-invasive adjustable frame design supports the home keyboards. Fourth, the design can be built with substantial reused content in the structure while still maintaining a clear architecture for future teams to build upon. Fifth, the design can be visually comprehended for an audience at an Engineering Expo.

The design also has some disadvantages compared to the customer requirements. The design requires more channels, wires, and power management compared to the moving carriage and linkage designs. Additionally, there is more emphasis on mechanical alignment between key stations.

5. Go-Forward Plan and Verification

5.1 Why This Design Is the Best Choice

The selected design is the best choice for this project because it meets the highest-priority requirements simultaneously without relying on speculative coordination strategies. The design also supports autonomous operation, supports playing more than one note at a time compared to playing just a single note, supports non-invasive attachment to a home keyboard, and supports a modular design for staged development. The design also draws on experience from prototype development, such as the move from relay to MOSFET switching.

5.2 Implementation Plan

Implementation must occur in well-defined stages: Stage 1: Validate one full actuation channel; Stage 2: Expand to a small group of adjacent keys; Stage 3: Integrate demonstration subset, hardware, user interface; Stage 4: Focus on repeatability testing, optimization, exterior design, and full song testing.

Table 3. Near-term implementation work packages aligned with the team workflow.

Work Package	Purpose	Near-Term Output
Mechanical mounting design	Create adjustable, non-invasive key-actuation geometry	CAD for solenoid mounts and keyboard interface
Electrical integration	Replace relay-based switching with proper high-speed driver hardware	MOSFET/PWM housing, wiring diagram, and safe power layout
Software pipeline cleanup	Enable reliable autonomous playback from song input to controller output	Bug-fixed code, organized storage, and streamlined Python-to-Arduino handoff
Validation and analysis	Retire risk before full integration	Single-key tests, small-cluster tests, and stress-analysis support

5.3 Verification Against Engineering Specifications

Verification must be directly related to specifications listed in Table 1. Autonomous reliability must be verified for 20 full songs with a definition of success as completion without human input from start. The song library requirement is satisfied when at least three songs can be selected for playback on demand. Setup must be verified for deployment from transport to playback ready within a timed period. System mass must also be verified to be less than 100 lb. Material must be verified to be 50% reused by part count to avoid assumption based on mass.

The quality of performance should be assessed by event-based testing and complete piece demonstrations. At the channel level, the team should ensure proper triggering, consistency, and thermal behavior during repeated playback. At the system level, the team should ensure that the keyboard is producing the right notes, that the mechanism is visually orderly and safe, and that

the sound is pleasing to the ear for public consumption.

6. Conclusion

Musical Bot Encore is a keyboard-playing robot designed to replace an aging outreach exhibit with a more reliable and extensible Engineering Expo demonstration. The design process began with stakeholder requirements, specification of these requirements, proof of concept to identify key risks, and comparing various system concepts based on weightings related to project goals.

The proposed design is not the lowest channel count design; rather, it is the strongest capstone design based on the requirements most critical to project success: autonomous operation, recognizable multi-note play, portability, support for various home keyboard types, and a path to future expansion.

The implementation plan is designed to retire hardware, software, and integration risks early in the project plan. If successfully executed, Musical Bot Encore will deliver an engaging and technically defensible exhibit to Oregon State University for future Engineering Expo attendees to engage with.

References

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